B1.2 Proteins

NAME	DATE	TOTAL
		_____ / 32

About 30 minutes, or one section at a time. A calculator is allowed. The number of marks shows how many separate points to make.

Section A: Recall

6 MARKS

1 STATE the name of the bond between amino acids. [B1.2.2] [1]

2 STATE the number of different amino acids coded for in the genetic code. [B1.2.4] [1]

3 DISTINGUISH between essential and non-essential amino acids. [B1.2.3] [2]

4 DEFINE denaturation. [B1.2.5] [1]

5 STATE the function of haemoglobin. [B1.2.4] [1]

Section B: Understanding and drawing

12 MARKS

6 DRAW a generalized amino acid, and label the groups. [B1.2.1] [3]

7 DRAW a dipeptide, labelling the peptide bond, and **write** the word equation for its formation. [B1.2.2] [3]

8 EXPLAIN why the possible variety of peptide chains has no real limit. [B1.2.4] [3]

9 EXPLAIN how a change in pH can stop a protein from functioning. [B1.2.5] [3]

Section C: Working with data

8 MARKS

All data in this section are invented for practice. The patterns are realistic.

10 [B1.2.5]

Students measured how long egg white took to turn solid at different temperatures, in tubes of the same size.

TEMPERATURE / °C	40	50	60	70	80
TIME TO TURN SOLID / S	No change	No change	900	120	30

a DESCRIBE the trend. [2]

b CALCULATE how many times faster denaturation is at 80°C than at 60°C. [1]

c EXPLAIN what is happening to the proteins as the egg white turns solid. [3]

d CALCULATE the number of possible sequences for a peptide of 5 amino acids. [2]

Section D: Longer answer

6 MARKS

11 **OUTLINE** the reasons why there are so many different proteins, each with a particular function. [6]

[B1.2.2, B1.2.4]